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# Posterior code across distributional shifts: zero-shot generalization and novel experimental design

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## 1 Intro

### 1.1 Background

- Can we experimentally leverage or test normative theory of NSC?
- Can we build data-driven, flexible, predictive models using NSC? [1]
- What fundamental advantage/main point do NSC models have in theory over system identification? Generative model instead of only discriminative model. Generative model consists of likelihood and prior and discriminative model consists only directly posterior, which involves knowledge of the underlying data distribution. What can we do about the data distribution?
- **Generalization across data distributions** Inference is data distribution (context) dependent, and so are neurons hypothesized to be. Can we make use of sampling's generative model? In theory, yes!
- **Generalization across tasks/context with the same data distribution?** More intuition needed

### 1.2 Philosophical/Axiomatic/Normative

- **Generative over discriminative** The brain maintains a generative model  $p(O, X_1, X_2, X_3, \dots)$  instead of only discriminative models  $p(X_1|O)$ ,  $p(X_2|O)$ , ...
- **Internal model vs external model** Brain maintains an internal, subjective model shaped by experience (phenomenological)
- **Posterior coding** Sensory areas encode marginal posteriors beliefs  $p(X_1|O)$ ,  $p(X_2|O)$ , ..., and in general the posterior  $p(X_1, X_2, X_3, \dots | O) \propto p(O|X_1)p(X_1|X_2)p(X_3|X_4), \dots$
- **Over long periods: Physical/causal mechanisms** Brain learns over long periods, the causal, generative mechanism (physics, conditional distribution) potentially over evolution short periods (tasks) learns anti-causal, inference mechanism (posterior)
- **Over short periods: Inference via prior** learns anti-causal, inference mechanism (posterior) by only learning the prior over context since posterior  $\propto$  causal mechanism  $\times$  prior
- **Invariance and independence of causal mechanisms** Causal mechanism is the physics, it does not change based on environments and is independent of tasks or contexts, making inference a problem of learning only the prior via stimulus statistics and/or task objective
- **Perception as a modular and efficient task** inference via fixed causal mechanism and prior learning

#### 1.2.1 Simulations

##### Simulation design

- Define true generative model learned by the brain (for both task 1 and task 2).
- Sample images and neuronal responses from both tasks.
- Train system identification model on data from one of the tasks.
- Learn generative model from one of the tasks and obtain posterior distribution.
- Compare performance of posterior vs system identification. It should match.
- Now change the tasks and then test the performance. Both models should suck.
- How to remedy system identification? Retrain the model on the new dataset.
- How to remedy generative model? Just learn prior purely from stimulus samples and recompute posterior  $\rightarrow$  zero shot generalization

### Generative models to consider

1. Simple Gaussian just to demonstrate the idea, see Section 3, Figure 1.
2. Ralf’s 2016 cardinal vs oblique orientation discrimination task [2]
3. Qamar’s 2013 task [3]

### 1.3 Data-driven

- Edgar R01 data
- Cohen and Newsome data? [4]
- Ralf V4 data?
- Diffusion model on V1/V4 data for single data distribution

## 2 Theory

In our previous work [1] we established that on a given dataset of neuronal responses  $\mathbf{r}$  and stimuli  $\mathbf{x}$ , Neural Sampling Code (NSC) generative models cannot surpass system identification models. We then contemplated a core theoretical strength of NSC: the ability to adapt to a changing stimulus distribution.

If we change the stimulus distribution  $p(\mathbf{x})$  to  $p_{\text{new}}(\mathbf{x})$  with markedly different stimulus statistics and let the sensory neurons adapt to  $p_{\text{new}}(\mathbf{x})$ , we would expect the system identification model’s performance to drop on  $p_{\text{new}}(\mathbf{x})$ . The system identification model would then have to be retrained on a new dataset of responses under  $p_{\text{new}}(\mathbf{x})$ . This is the case where we would expect the NSC’s learned generative model to be beneficial. Specifically, change in  $p(\mathbf{x})$  to  $p_{\text{new}}(\mathbf{x})$  may entirely derive from the change in prior  $p(\mathbf{z})$  to  $p_{\text{new}}(\mathbf{z})$ , while  $p(\mathbf{x}|\mathbf{z})$  remains fixed. Consequently, if NSC accurately describes a neural population, i.e.,  $\mathbf{r} \sim p(\mathbf{z}|\mathbf{x}) \propto p(\mathbf{x}|\mathbf{z})p(\mathbf{z})$ , the neuronal adaptation can be accounted for by simply learning  $p_{\text{new}}(\mathbf{z})$ , i.e.,  $\mathbf{r}_{\text{new}} \sim p_{\text{new}}(\mathbf{z}|\mathbf{x}) \propto p(\mathbf{x}|\mathbf{z})p_{\text{new}}(\mathbf{z})$ , keeping  $p(\mathbf{x}|\mathbf{z})$  fixed.

**Task transfer** We aim to realize the change in stimulus statistics by introducing “tasks”, where each task modulates the stimulus distribution via the following graphical model:  $\mathbf{t} \rightarrow \mathbf{z} \rightarrow \mathbf{x}$ . Considering two tasks  $\mathbf{t} = A$  and  $\mathbf{t} = B$ , the posteriors can be written as  $p(\mathbf{z}|\mathbf{x}, \mathbf{t} = A) \propto p(\mathbf{x}|\mathbf{z})p(\mathbf{z}|\mathbf{t} = A)$  and  $p(\mathbf{z}|\mathbf{x}, \mathbf{t} = B) \propto p(\mathbf{x}|\mathbf{z})p(\mathbf{z}|\mathbf{t} = B)$ . Importantly, we showed in our previous work [1] that the generative model (for a given task),  $p(\mathbf{x}, \mathbf{z}|\mathbf{t}) = p(\mathbf{x}|\mathbf{z})p(\mathbf{z}|\mathbf{t})$  can be learned via a dataset of neuronal responses and stimuli recorded for that task and discussed the theoretical possibility of “transferring” learned likelihood  $p(\mathbf{x}|\mathbf{z})$  across tasks (stimulus distributions), and *adapting* the learned generative model by just learning the prior of the new task:

$$\arg \max_{\theta} p(\{\mathbf{x}^{(i)}\}_{i=1}^N | \mathbf{t}_{\text{new}}) = \arg \max_{\theta} \int p(\{\mathbf{x}^{(i)}\}_{i=1}^N | \mathbf{z}) p(\mathbf{z} | \mathbf{t}_{\text{new}}; \theta), \quad (1)$$

where  $\{\mathbf{x}^{(i)}\}_{i=1}^N \sim p(\mathbf{x}|\mathbf{t}_{\text{new}})$ . Crucially, note that in order to learn the new generative model, we only had to adapt the prior. Given the structure of the generative model, adapting the new prior only requires stimulus samples and no samples of neuronal responses (latent variables), thus providing *zero-shot generalization*.

### 3 Simulations

#### 3.1 Full Gaussian “strawman” model

We assume a simple generative model where the posterior is analytically tractable:

$$p(\mathbf{t}) = \text{Bernoulli}(p_t) \quad (2)$$

$$p(\mathbf{z}|\mathbf{t}) = \mathcal{N}(\mathbf{z}|\mu_t, \Sigma_t) \quad (3)$$

$$p(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}|\mathbf{Q}\mathbf{z} + \mathbf{w}, \mathbf{L}) \quad (4)$$

The task conditioned posterior over the latent variable,  $\mathbf{z}$  (task conditioned neuronal response distribution) is given by:

$$p(\mathbf{z}|\mathbf{x}, \mathbf{t}) = \mathcal{N}(\mathbf{z}|\mathbf{C}\{\mathbf{Q}^T\mathbf{L}(\mathbf{x} - \mathbf{w}) + \Sigma^{-1}\mu_t\}, \mathbf{C}) \quad (5)$$

where  $\mathbf{C} = (\Sigma^{-1} + \mathbf{Q}^T\mathbf{L}^{-1}\mathbf{Q})^{-1}$ .

We assume two tasks, a source task,  $\mathbf{t} = A$  and a target task  $\mathbf{t} = B$  and between the two tasks we only change the priors  $p(\mathbf{z}|\mathbf{t})$  (Figure 1a) and preserve the likelihood  $p(\mathbf{x}|\mathbf{z})$ . We assume that the brain can learn these generative models perfectly and therefore they also represent the true generative models maintained by the brain.

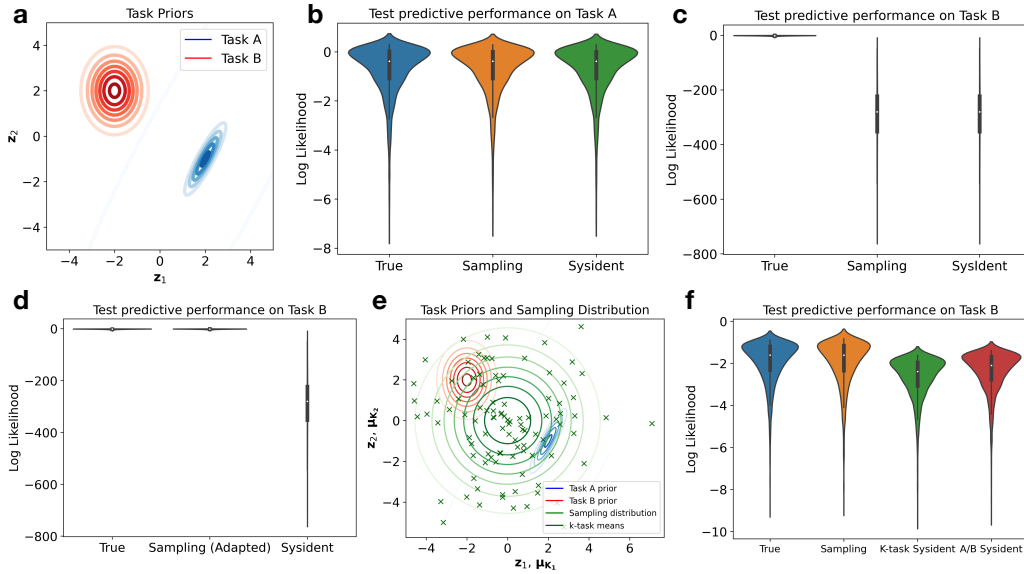


Figure 1: Task transfer. **a** Prior distributions for tasks A and B. **b** Predictive performance of NSC posterior (Sampling) and system identification model (Sysident) on testset of task A, with both trained on trainset of task A, in comparison to true log-likelihood. **c** Predictive performance of NSC posterior and system identification model on testset of task B, with both trained on trainset of task A, in comparison to true log-likelihood. **d** Predictive performance of *adapted* NSC posterior (Sampling (Adapted)) and system identification model on testset of task B, with system identification model trained on task A, in comparison to true log-likelihood. **e** Task A and Task B prior distributions, k-task mean distribution. **f** Predictive performance of *adapted* NSC posterior (Sampling (Adapted)), system identification model trained on k-tasks (K-task Sysident), and system identification model trained on combined dataset of tasks A and B (A/B sysident), in comparison to true log-likelihood.

We then sample a dataset of  $N = 10,000$  pairs of neuronal responses (latent variable samples,  $\mathbf{z}$ ) and images  $\mathbf{x}$  from  $\mathbf{t} = A$ , i.e.,  $\{\mathbf{z}^{(i)}, \mathbf{x}^{(i)}\}_{i=1}^N \sim p(\mathbf{z}, \mathbf{x}|\mathbf{t} = A)$ . We first learn the NSC generative model on a train split (0.8x) of the dataset using the method we describe in our previous work [1]. We then invert the learned generative model by computing the posterior analytically. The posterior  $p(\mathbf{z}|\mathbf{x}, \mathbf{t} = A)$  represents a neuronal response predictive model under task A. Additionally, we fit

a Gaussian system identification model from the training set of task A. As expected, the posterior as well as the system identification model learn the true posterior of the task generative model (Figure 1b).

We then sample a dataset from task B and check the generalizability of the NSC posterior as well as the system identification and see that their performance suffers in the new task (Figure 1c). *Given our theory, we adapt the learned generative model to task B by just learning the prior of the task B's stimulus distribution without the need for neuronal responses, and then find its posterior.* We clearly see that the posterior from the adapted model recovers the underlying true posterior resulting in *zero-shot generalization* (Figure 1d).

Furthermore, in order to aid the system identification model further, we train it on  $K = 100$  tasks (Figure 1e). Note that these tasks do *not* include the target task B. We additionally train another system identification model which is trained on the combined training sets of both tasks A and B. We clearly see that both these conditions significantly aid the system identification model resulting in a strong boost in its performance (Figure 1f). Crucially though, the system identification models still do not match the performance of the adapted NSC model. This theoretically demonstrates our prediction from the normative theory of NSC: its adaptability to different stimulus distributions allowing zero-shot generalization.

### 3.2 Orientation discrimination task models

Can we show that the same holds for (a) more realistic models of the brain and (b) under more realistic tasks contexts? Here we turn towards orientation discrimination tasks and established models for early sensory areas from the literature. Following Qamar et al. [3] and Haefner, Berkes, and Fiser [2], we assume the following task and brain generative models (Figure 2).

**Generative model** See Figure 2

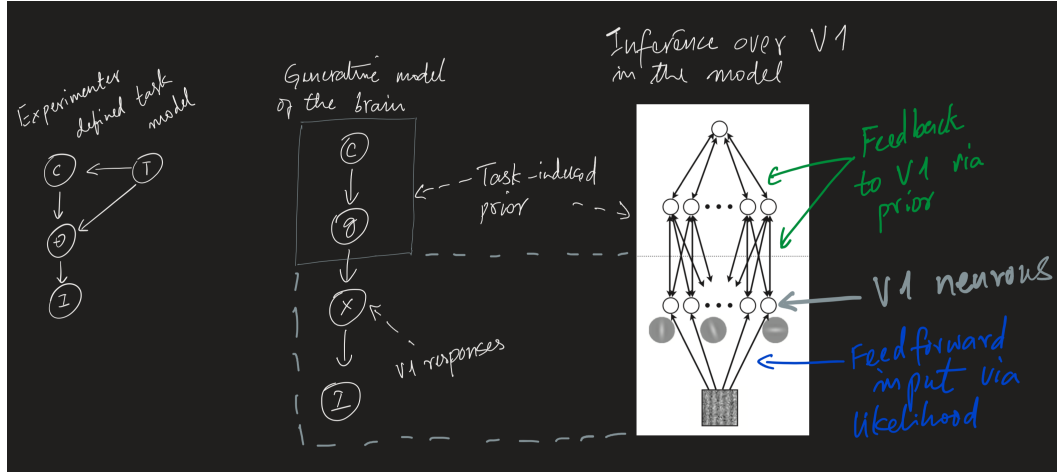


Figure 2: Experimenter defined and (assumed) generative model of the brain. Model and illustration borrowed from Haefner, Berkes, and Fiser [2].

The sensory variables in this model are  $\mathbf{x}$  and the brain is hypothesized to compute the posterior  $p(C, g_1, g_2, \dots, g_{n_g}, x_1, x_2, \dots, x_{n_x} | I, T)$ .

The generative model implies the joint distribution:

$$p(C, g_1, g_2, \dots, g_{n_g}, x_1, x_2, \dots, x_{n_x}, I, T) = p(C | T) \prod_{i=1}^{n_g} p(g_i | C, T) \cdot \prod_{j=1}^{n_x} p(x_j | g) \cdot p(I | x)$$

Posterior in this model, unlike in the Gaussian case are intractable, but we can approximate it via Gibbs sampling, as in Haefner, Berkes, and Fiser [2]:

**Posterior inference via Gibbs sampling** Posterior:  $p(C, g_1, g_2, \dots, g_{n_g}, x_1, x_2, \dots, x_{n_x} \mid I, T)$

Gibbs sampling algorithm:

1. Initialize vector  $\{C^{(1)}, g_1^{(1)}, g_2^{(1)}, \dots, g_{n_g}^{(1)}, x_1^{(1)}, x_2^{(1)}, \dots, x_{n_x}^{(1)}\}$
2. For  $t = 1, \dots, \tau$ 
  - Sample  $C^{t+1} \sim p(C \mid g_1^{(t)}, g_2^{(t)}, \dots, g_{n_g}^{(t)}, x_1^{(t)}, x_2^{(t)}, \dots, x_{n_x}^{(t)}, I, T)$
  - Sample  $g_1^{t+1} \sim p(g_1 \mid g_2^{(t)}, g_3^{(t)}, \dots, g_{n_g}^{(t)}, x_1^{(t)}, x_2^{(t)}, \dots, x_{n_x}^{(t)}, C^{(t)}, I, T)$
  - Sample  $g_2^{t+1} \sim p(g_2 \mid g_1^{(t)}, g_3^{(t)}, \dots, g_{n_g}^{(t)}, x_1^{(t)}, x_2^{(t)}, \dots, x_{n_x}^{(t)}, C^{(t)}, I, T)$
  - ...
  - Sample  $g_{n_g}^{t+1} \sim p(g_{n_g} \mid g_1^{(t)}, g_2^{(t)}, \dots, g_{n_g-1}^{(t)}, x_1^{(t)}, x_2^{(t)}, \dots, x_{n_x}^{(t)}, C^{(t)}, I, T)$
  - Sample  $x_1^{t+1} \sim p(x_1 \mid g_1^{(t)}, g_2^{(t)}, \dots, g_{n_g}^{(t)}, x_2^{(t)}, \dots, x_{n_x}^{(t)}, C^{(t)}, I, T)$
  - ...
  - Sample  $x_{n_x}^{t+1} \sim p(x_{n_x} \mid g_1^{(t)}, g_2^{(t)}, \dots, g_{n_g}^{(t)}, x_1^{(t)}, \dots, x_{n_x}^{(t)}, C^{(t)}, I, T)$

Hence, corresponding Gibbs posteriors are as follows:

$$p(C \mid g_1, g_2, \dots, g_{n_g}, x_1, x_2, \dots, x_{n_x}, I, T)$$

$$p(g_i \mid \{g_j\}_{j=1, j \neq i}^{n_g}, x_1, x_2, \dots, x_{n_x}, C, I, T)$$

$$p(x_i \mid g_1, g_2, \dots, g_{n_g}, \{x_j\}_{j=1, j \neq i}^{n_g}, C, I, T)$$

In order to derive the Gibbs posteriors, we can first simplify the joint distribution via conditional independencies as follows:

$$p(C, g_1, g_2, \dots, g_{n_g}, x_1, x_2, \dots, x_{n_x}, I, T) = p(C \mid T) \prod_{i=1}^{n_g} p(g_i \mid C, T) \cdot \prod_{j=1}^{n_x} p(x_j \mid g) \cdot p(I \mid x)$$

**Learning the generative model** Having recorded a dataset of sensory stimuli and corresponding sensory neuronal responses, denoted  $\{I^{(i)}, x^{(i)}\}$  (which we can simulate here by simply sampling from the above generative model under one task), we can learn the generative model using our framework [1] as:

$$\theta^* = \arg \max_{\theta} \sum_{i=1}^N p(\mathbf{I}^{(i)}, \mathbf{x}^{(i)} \mid T; \theta) \tag{6}$$

$$\theta_L^*, \theta_P^* = \arg \max_{\theta_L, \theta_P} \sum_{i=1}^N p(\mathbf{I}^{(i)} \mid \mathbf{x}^{(i)}; \theta_L) p(\mathbf{x}^{(i)} \mid T; \theta_P)$$

If we knew the right structure of the hierarchical prior, we could write out the prior model further to take on the known structure:

$$\theta_L^*, \theta_P^* = \arg \max_{\theta_L, \theta_P} \sum_{i=1}^N p(\mathbf{I}^{(i)} \mid \mathbf{x}^{(i)}; \theta_L) p(\mathbf{x}^{(i)} \mid T; \theta_P)$$

$$\theta_L^*, \theta_{\mathbf{x}}^*, \theta_{\mathbf{g}}^*, \theta_C^* = \arg \max_{\theta_L, \theta_{\mathbf{x}}, \theta_{\mathbf{g}}, \theta_C} \sum_{i=1}^N p(\mathbf{I}^{(i)} \mid \mathbf{x}^{(i)}; \theta_L) \sum_C \sum_{\mathbf{g}} p(\mathbf{x} \mid \mathbf{g}; \theta_{\mathbf{x}}) p(\mathbf{g} \mid C, T; \theta_{\mathbf{g}}) p(C \mid T; \theta_C) \tag{7}$$

Above only works when (1)  $\mathbf{g}$  is low-dimensional, since otherwise summing over all possible  $\mathbf{g}$ -vectors is going to be a combinatorial explosion, and (2) we know the right hierarchical structure of the prior. In general, neither of them is true. Hence, we can in general learn the prior flexibly using a normalizing flow.

**Learning the posterior via variational inference** We can approximate the true posterior  $p(\mathbf{x} | \mathbf{I}; \theta^*)$  by fitting a model posterior  $q(\mathbf{x} | \mathbf{I}, T; \phi)$  to samples from  $p(\mathbf{I}, \mathbf{x} | T; \theta^*)$  directly via maximum log-likelihood:

$$\phi^* = \arg \max_{\phi} \sum_{\mathbf{I}', \mathbf{x}'} \log q(\mathbf{x}' | \mathbf{I}', T; \phi), \quad (8)$$

where  $\mathbf{I}', \mathbf{x}' \sim p(\mathbf{I}, \mathbf{x} | T; \theta^*)$ , are samples from the trained generative model.

**Across task context generalization** For  $T = T'$ , we ought to learn the new prior via stimulus samples only and if (1) and (2) above were true:

$$\begin{aligned} \arg \max_{\theta'^*} p(\mathbf{I} | T') &= \arg \max_{\theta'} \sum_C \sum_g \int_z p(\mathbf{I}, \mathbf{x}, \mathbf{g}, C | T; \theta') d\mathbf{z} \\ &= \arg \max_{\theta'} \sum_C \sum_g \int_z p(\mathbf{I} | \mathbf{x}; \theta_L^*) p(\mathbf{x} | \mathbf{g}; \theta') p(\mathbf{g} | C, T') p(C | T') d\mathbf{x} \end{aligned} \quad (9)$$

Otherwise (general case):

$$\begin{aligned} \arg \max_{\theta'^*} p(\mathbf{I} | T') &= \arg \max_{\theta'} \int_z p(\mathbf{I}, \mathbf{x}, | T'; \theta') d\mathbf{z} \\ &= \arg \max_{\theta'} \int_z p(\mathbf{I} | \mathbf{x}; \theta_L^*) p(\mathbf{x}; \theta') d\mathbf{x} \end{aligned} \quad (10)$$

**Adapting model posterior via new prior** We can now learn an adapted posterior for task context  $T'$ ,  $q(\mathbf{x}' | \mathbf{I}', T'; \phi'^*)$  via the learned prior:

$$\phi'^* = \arg \max_{\phi'} \sum_{\mathbf{I}', \mathbf{x}'} \log q(\mathbf{x}' | \mathbf{I}', T; \phi'), \quad (11)$$

where  $\mathbf{I}', \mathbf{x}' \sim p(\mathbf{I}, \mathbf{x} | T; \theta^*, \theta'^*)$ , are samples from the *adapted* generative model (new prior).

**Learning system identification model** Straight-up maximum likelihood on recorded data:

$$\theta_{\text{SI}}^* = \arg \max_{\theta_{\text{SI}}} \sum_{i=1}^N \log q(\mathbf{x}^{(i)} | \mathbf{I}^{(i)}, T; \theta_{\text{SI}}) \quad (12)$$

In order to facilitate a fair comparison, it is crucial that the form of the model density  $q$  (and parameter/amortization function) between system identification and NSC posterior match exactly.

We perform the above steps for two specific orientation discrimination models.

### 3.2.1 Haefner 2016 model

**Task** We follow the task from Cohen and Newsome [4] Figure 3.

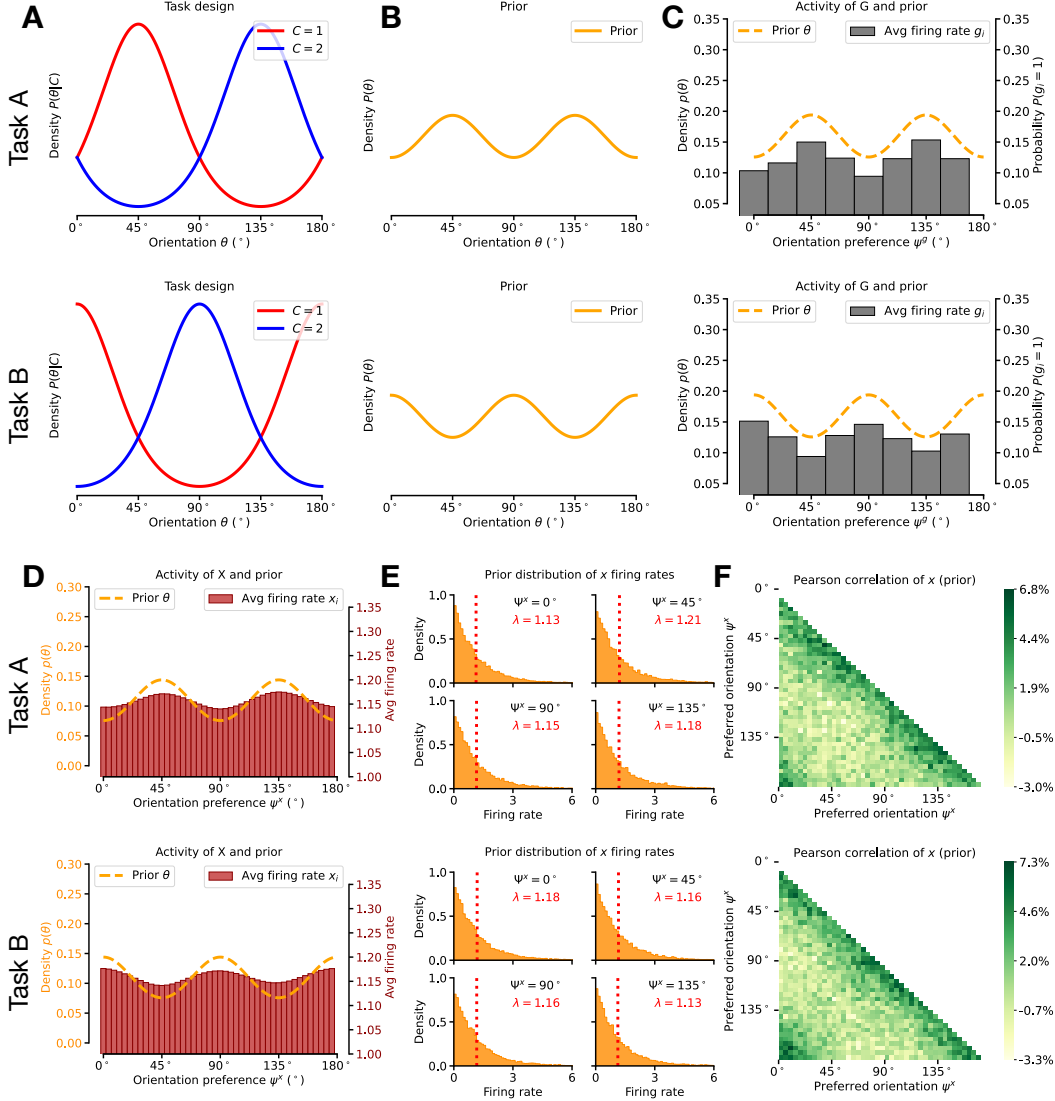


Figure 3: **A, B** Orientation discrimination task as in Cohen and Newsome [4], **C—F** How orientation discrimination task modulates activity of  $G$  and  $X$  neurons. **TODO: detailed description.**

### Generative model

$$p(C | T) = \text{Bernoulli}(p^{(T)})$$

$$p(g_i | C, T) = \text{Bernoulli} \left( p_{i,C,T}^{(g)} = \frac{1}{n_g I_0(\kappa)} \exp \left( \kappa \cos 2(\theta_i^{(g)} - \theta_{C,T}) \right) \right)$$

$$p(x_i | g) = \frac{1}{\lambda_i} \exp \frac{-x_i}{\lambda_i} H(x_i)$$

$$\lambda_i = E[x_i | g] = 1 + \delta \sum_{k=1}^{n_g} g_k \exp \left[ \lambda \cos 2 \left( \theta_i^{(x)} - \theta_k^{(g)} \right) \right]$$

$$P(I | x) = \mathcal{N}(I | Ax, \sigma_I^2)$$

The task modulates activity of  $G$  as well as  $X$  neurons differently Figure 3.

### 3.2.2 Qamar 2013 model (novel)

Same generative model as above can be reused.

## 4 Experiments on recorded data

Repeat learning procedure exactly as in simulation for any recorded data.

### 4.1 Edgars R01 data

Issue: no significant difference found between populations, tested via decoding approach (for approach details, see Walker et al. [5]).

Hypothesis: task priors were not widely separated enough.

It is possible that via our posterior model approach we do see (in theory and in practice) a difference. If so then win, our zero shot prediction should work well! If not, several possibilities: models not good enough, likelihood transfer assumptions doesn't hold, animal learned the wrong prior, the priors were indeed not separate enough. How do we rule out everything other than the last hypothesis? If we manage to, then we can sell our framework as a guide to do future experiments: separate the priors further and you'll see the effect.

### 4.2 Task free data (V1, V4, ...)

1. Revisit Fabee's idea of extrapolating prior from natural to artificial

$$p_{\text{Natural}}(\mathbf{x})^{1-\alpha} \cdot p_{\text{Ideal-observer}}(\mathbf{x}|T)^\alpha \quad (13)$$

2. Diffusion model based generative model

## 5 Experimental design

Information theoretic optimization criterion to design priors.

## 6 Related works

### 6.1 Experimental underpinnings of task/contextual modulation

Lange et al. [6], Cohen and Newsome [4]

### 6.2 Theory papers exploring task/contextual modulation

Haefner, Berkes, and Fiser [2], Lange and Haefner [7]



### 6.3 Zero-shot/k-shot system identification

## 7 Extra

### 7.1 Binary neuron model

#### Generative model

$$\begin{aligned}
p(C | T) &= \text{Bernoulli}(p^{(T)}) \\
p(g_i | C, T) &= \text{Bernoulli} \left( p_{i,C,T}^{(g)} = \exp \left( -\frac{1}{2\sigma_{C,T}^2} (\theta_i^{(g)} - \theta_{C,T})^2 \right) \right) \\
p(x_i | g) &= \text{Bernoulli} \left( p_i^{(x)} = \frac{1}{n_g} \sum_{j=1}^{n_g} \exp \left( -\frac{1}{2\sigma_x^2} (\theta_i^{(z)} - \theta_j^{(z)})^2 \right) \right) \\
P(I | x) &= \mathcal{N}(I | Ax, \sigma_I^2)
\end{aligned}$$

**Posterior inference via Gibbs sampling** Posterior:  $P(C, g_1, g_2, \dots, g_{n_g}, x_1, x_2, \dots, x_{n_x} | I, T)$

Correspondingly, the Gibbs posteriors can be simplified as follows:

$$\begin{aligned}
p(c | g_1, g_2, \dots, g_{n_g}, x_1, x_2, \dots, x_{n_x}, I, T) &= \frac{p(C | T) \prod_{i=1}^{n_g} p(g_i | C, T)}{\sum_{C \in \{0,1\}} p(C | T) \prod_{i=1}^{n_g} p(g_i | C, T)} \\
p(g_i | \{g_j\}_{j=1, j \neq i}^{n_g}, x_1, x_2, \dots, x_{n_x}, C, I, T) &= \frac{\prod_{j=1}^{n_g} p(g_j | C, T) \prod_{j=1}^{n_x} p(x_j | g)}{\sum_{g_i \in \{0,1\}} p(g_i | C) \prod_{j=1, j \neq i}^{n_g} p(g_j | C, T) \prod_{i=1}^{n_x} p(x_i | g)} \\
p(x_i | g_1, g_2, \dots, g_{n_g}, \{x_j\}_{j=1, j \neq i}^{n_g}, C, I, T) &= \frac{\prod_{j=1}^{n_x} p(x_j | g) \cdot p(I | x)}{\sum_{x_i \in \{0,1\}} p(x_i | g) \prod_{j=1, j \neq i}^{n_x} p(x_j | g) \cdot p(I | x)}
\end{aligned}$$

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